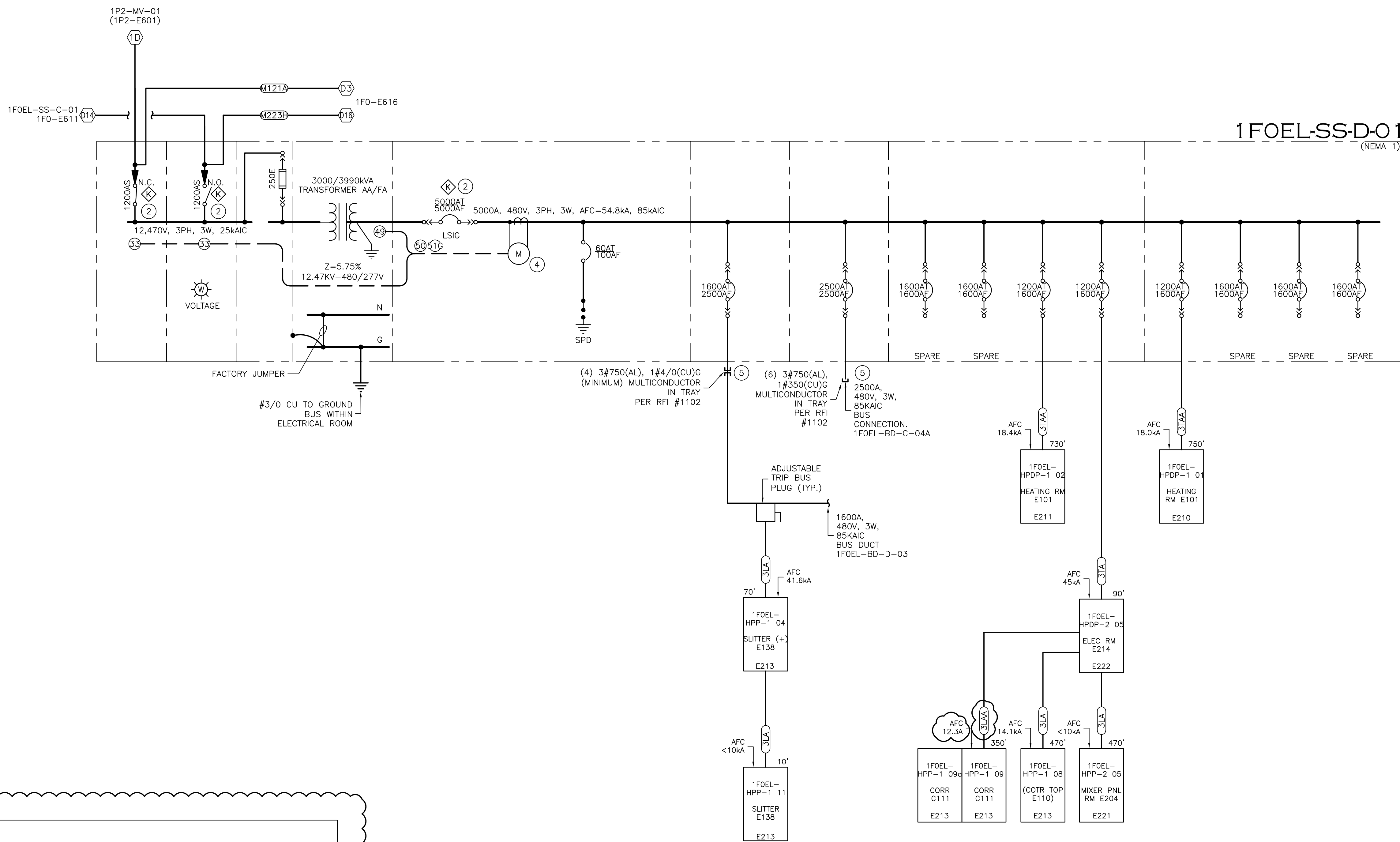




## GENERAL NOTES

- A. REFERENCE FEEDER SCHEDULE, E011
- B. CABLE LENGTHS WHEN INDICATED ON ONE-LINE ARE APPROXIMATE AND ARE FOR REFERENCE ONLY FOR CALCULATIONS AND ARE NOT TO BE USED FOR MATERIAL TAKEOFFS.
- C. SECONDARY UNIT SUBSTATIONS: REFER TO SPECIFICATION #26 11 16 12.
- D. UNIT SUBSTATION SCADA COMMUNICATIONS SHALL INCLUDE ALL HARDWARE DEVICES TO COMMUNICATE I/O (VIA MODBUS TCP COMMUNICATIONS) INCLUDING, BUT NOT LIMITED TO, THE FOLLOWING:
- PRIMARY SWITCH STATUS (OPEN/CLOSED)
  - TRANSFORMER WINDING TEMPERATURE
  - TRANSFORMER FAN RUN STATUS
  - BREAKER STATUS
  - POWER METER DATA
- E. ALL SWITCHBOARDS, SWITCHGEAR, AND PANELBOARDS SUPPLIED BY FEEDERS SHALL BE PERMANENTLY MARKED TO INDICATE EACH DEVICE OR EQUIPMENT WHERE THE POWER ORIGINATES. THE LABEL SHALL BE PERMANENTLY AFFIXED OF SUFFICIENT DURABILITY TO WITHSTAND THE ENVIRONMENT INVOLVED, AND NOT HANDWRITTEN.

KEYED NOTES 

1. NOT USED
2. PRIOR TO CLOSING A SWITCH, THE OTHER SWITCH MUST BE OPENED USING KEY TO INTERLOCK TO PREVENT CONNECTION OF MULTIPLE SOURCES. REFER TO SEQUENCE OF OPERATIONS BY MANUFACTURER IN EQUIPMENT SUBMITTAL.
3. NOT USED.
4. SQUARE D/ SCHNEIDER ION 9000 WITH RS-485 COMMUNICATIONS.
5. ENGINEERED CABLE BUS ASSEMBLY, CONDUIT, OR CABLE TRAY TO BUS DUCT CABLE TAP BOX.

| BUS DUCT NAME |                                   |           |       |       |           |          |               |          |          |       |
|---------------|-----------------------------------|-----------|-------|-------|-----------|----------|---------------|----------|----------|-------|
| 1FOEL-BD-C04A |                                   |           |       |       |           |          |               |          |          |       |
| BUS PLUG      |                                   | RATINGS   |       |       |           |          | LOADING (KVA) |          |          | NOTES |
|               |                                   | EQUIPMENT | TAG # | VOLTS | AMP FRAME | AMP TRIP | LOAD          | DEM LOAD | DEM LOAD |       |
| CATHODE(+)    | AC3A POWER PANEL                  | 1         | 480   | 250   | 250       | 165      | 0.4           | 66       | 198      |       |
| CATHODE(+)    | AC3B POWER PANEL                  | 2         | 480   | 250   | 250       | 176      | 0.4           | 70       | 212      |       |
| CATHODE(+)    | AC4A POWER PANEL                  | 3         | 480   | 250   | 250       | 165      | 0.4           | 66       | 198      |       |
| CATHODE(+)    | AC4B POWER PANEL                  | 4         | 480   | 250   | 250       | 176      | 0.4           | 70       | 212      |       |
| CATHODE(+)    | AC5 POWER PANEL                   | 5         | 480   | 250   | 250       | 141      | 0.4           | 57       | 170      |       |
| CATHODE(+)    | AC6 POWER PANEL                   | 6         | 480   | 400   | 400       | 270      | 0.4           | 108      | 325      |       |
| CATHODE(+)    | AC7 POWER PANEL                   | 7         | 480   | 400   | 400       | 270      | 0.4           | 108      | 325      |       |
| CATHODE(+)    | COATER PUA PANEL                  | 8         | 480   | 150   | 125       | 82       | 0.4           | 33       | 99       |       |
| CATHODE(+)    | COATER PUB PANEL                  | 9         | 480   | 150   | 50        | 35       | 0.4           | 14       | 42       |       |
| CATHODE(+)    | AC3A POWER PANEL                  | 10        | 480   | 250   | 250       | 165      | 0.4           | 66       | 198      |       |
| CATHODE(+)    | AC3B POWER PANEL                  | 11        | 480   | 250   | 250       | 176      | 0.4           | 70       | 212      |       |
| CATHODE(+)    | AC4A POWER PANEL                  | 12        | 480   | 250   | 250       | 165      | 0.4           | 66       | 198      |       |
| CATHODE(+)    | AC4B POWER PANEL                  | 13        | 480   | 250   | 250       | 176      | 0.4           | 70       | 212      |       |
| CATHODE(+)    | AC5 POWER PANEL                   | 14        | 480   | 250   | 250       | 141      | 0.4           | 57       | 170      |       |
| CATHODE(+)    | AC6 POWER PANEL                   | 15        | 480   | 400   | 400       | 270      | 0.4           | 108      | 325      |       |
| CATHODE(+)    | AC7 POWER PANEL                   | 16        | 480   | 400   | 400       | 270      | 0.4           | 108      | 325      |       |
| CATHODE(+)    | COATER PUA PANEL                  | 17        | 480   | 150   | 125       | 82       | 0.4           | 33       | 99       |       |
| CATHODE(+)    | COATER PUB PANEL                  | 18        | 480   | 150   | 50        | 35       | 0.4           | 14       | 42       |       |
| CATHODE(+)    | NO.3 BACK SLURRY TRANSFER PANEL   | 19        | 480   | 150   | 100       | 59       | 0.4           | 24       | 71       |       |
| CATHODE(+)    | FEED VESSEL WH-C31V1(300L) TOP    | 20        | 480   | 250   | 150       | 51       | 0.4           | 20       | 61       |       |
| CATHODE(+)    | FEED VESSEL WH-C32V1(300L) BACK   | 21        | 480   | 250   | 150       | 51       | 0.4           | 20       | 61       |       |
| CATHODE(+)    | NO.3 TOP SLURRY TRANSFER PANEL    | 22        | 480   | 150   | 100       | 59       | 0.4           | 24       | 71       |       |
| CATHODE(+)    | TRANSFER TANK WH-C31W1(800L) TOP  | 23        | 480   | 250   | 150       | 59       | 0.4           | 24       | 71       |       |
| CATHODE(+)    | TRANSFER TANK WH-C32W1(800L) BACK | 25        | 480   | 250   | 150       | 59       | 0.4           | 24       | 71       |       |
| CATHODE(+)    | TRANSFER TANK WH-C22W1(800L) BACK | 27        | 480   | 250   | 150       | 59       | 0.4           | 24       | 71       |       |
| CATHODE(+)    | FEED VESSEL WH-C22V1(300L) BACK   | 29        | 480   | 250   | 150       | 59       | 0.4           | 24       | 71       |       |
| CATHODE(+)    | COATER UPS                        | 30        | 480   | 800   | 800       | 555      | 0.4           | 222      | 668      | *     |

NOTES:

\*800A AND ABOVE BUS PLUGS REQUIRE MANUFACTURER INSTALLED BUS PLATE. SEE POWER PLAN FOR LOCATION

TOTAL LOAD (KVA)

397.3

1589

TOTAL DEMAND LOAD (KVA)

1911.4 DEMAND AMPS

2500 RATING (A)

76 % LOADING

1 ONE LINE DIAGRAM - (1 FO) ELECTRODE "D" (PARTIAL)  
NO SCALE

| BUS DUCT NAME                                                                                   |                        |                  |       |       |     |      |               |     |                             |       |
|-------------------------------------------------------------------------------------------------|------------------------|------------------|-------|-------|-----|------|---------------|-----|-----------------------------|-------|
| 1FOEL-BD-D03                                                                                    |                        |                  |       |       |     |      |               |     |                             |       |
| BUS PLUG                                                                                        |                        | RATINGS          |       |       |     |      | LOADING (KVA) |     |                             | NOTES |
|                                                                                                 |                        | TAG #            | VOLTS | FRAME | AMP | LOAD | DEM           | FLA |                             |       |
| CATHODE(+)                                                                                      | MAIN PANEL CHILLER #1  | 1                | 480   | 250   | 200 | 141  | 0.4           | 57  | 170                         |       |
| CATHODE(+)                                                                                      | MAIN PANEL CHILLER #2  | 2                | 480   | 250   | 200 | 141  | 0.4           | 57  | 170                         |       |
| CATHODE(+)                                                                                      | MAIN PANEL CHILLER #3  | 3                | 480   | 250   | 200 | 141  | 0.4           | 57  | 170                         |       |
| CATHODE(+)                                                                                      | MAIN PANEL CHILLER #4  | 4                | 480   | 250   | 200 | 141  | 0.4           | 57  | 170                         |       |
| CATHODE(+)                                                                                      | MAIN PANEL CHILLER #5  | 5                | 480   | 250   | 200 | 141  | 0.4           | 57  | 170                         |       |
| CATHODE(+)                                                                                      | MAIN PANEL CHILLER #6  | 6                | 480   | 250   | 200 | 141  | 0.4           | 57  | 170                         |       |
| CATHODE(+)                                                                                      | MAIN PANEL CHILLER #7  | 7                | 480   | 250   | 200 | 141  | 0.4           | 57  | 170                         |       |
| CATHODE(+)                                                                                      | MAIN PANEL CHILLER #8  | 8                | 480   | 250   | 200 | 141  | 0.4           | 57  | 170                         |       |
| CATHODE(+)                                                                                      | MAIN PANEL CHILLER #9  | 9                | 480   | 250   | 200 | 141  | 0.4           | 57  | 170                         |       |
| CATHODE(+)                                                                                      | MAIN PANEL CHILLER #10 | 10               | 480   | 250   | 200 | 141  | 0.4           | 57  | 170                         |       |
| CATHODE(+)                                                                                      | MAIN PANEL CHILLER #11 | 11               | 480   | 250   | 200 | 141  | 0.4           | 57  | 170                         |       |
| CATHODE(+)                                                                                      | MAIN PANEL CHILLER #12 | 12               | 480   | 250   | 200 | 141  | 0.4           | 57  | 170                         |       |
| 1FE1-HDP-1-04                                                                                   |                        | 13               | 480   | 250   | 200 | 58   | 1.0           | 58  | 70                          |       |
| PANCake REWINDER CATHODE IHU PANEL                                                              |                        | 14               | 480   | 250   | 100 | 59   | 0.4           | 24  | 71                          |       |
| PANCake REWINDER CATHODE IHU PANEL                                                              |                        | 15               | 480   | 250   | 100 | 59   | 0.4           | 24  | 71                          |       |
| PANCake REWINDER CATHODE IHU PANEL                                                              |                        | 16               | 480   | 250   | 100 | 58   | 0.4           | 24  | 71                          |       |
| NOTES                                                                                           |                        |                  |       |       |     |      |               |     |                             |       |
| *800A AND ABOVE BUS PLUGS REQUIRE MANUFACTURER INSTALLED BUS PLATE. SEE POWER PLAN FOR LOCATION |                        | TOTAL LOAD (KVA) |       |       |     |      | 1931          |     | 807 TOTAL DEMAND LOAD (KVA) |       |
|                                                                                                 |                        |                  |       |       |     |      |               |     | 970.918 TOTAL DEMAND AMPS   |       |
|                                                                                                 |                        |                  |       |       |     |      |               |     | 1600 RATING (A)             |       |
|                                                                                                 |                        |                  |       |       |     |      |               |     | 61% LOADING                 |       |

### 3 BUS DUCT LOAD SCHEDULE

|          |          |            |             |            |                            |     |     |     |     |
|----------|----------|------------|-------------|------------|----------------------------|-----|-----|-----|-----|
| INCOMING | INCOMING | TRANSITION | TRANSFORMER | TRANSITION | MAIN<br>CIRCUIT<br>BREAKER | ACB | ACB | ACB | ACB |
|          |          |            |             |            |                            |     |     | ACB | ACB |
|          |          |            |             |            |                            |     |     | ACB | ACB |
|          |          |            |             |            |                            |     |     | ACB | ACB |

## 2 EQUIPMENT ELEVATION

| DATE       | REVISION                         |
|------------|----------------------------------|
| 06/26/2024 | 1F0 - 2170 REVISIONS             |
| 06/28/2024 | PERMIT RESUBMISSION B23-440      |
| 10/16/2024 | 1DC-074 1F0 CD PROGRESS SET (ME) |
| 11/10/2024 | 1DC-074 1F0 FULL CD PACKAGE      |
| 12/17/2024 | 1F0 1DC-074 REVISIONS            |
| 03/11/2025 | 1F0 CD REVISIONS                 |
| 06/10/2025 | 1F0 CD RESUBMISSION              |

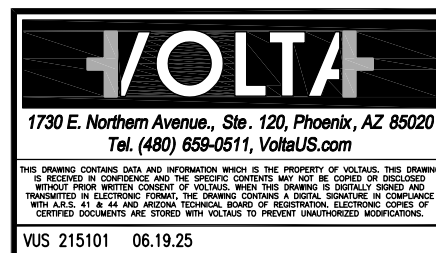
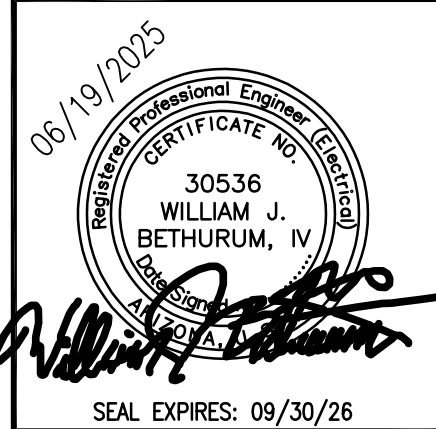
|                   |               |
|-------------------|---------------|
| <b>PA/PM:</b>     | WB            |
| <b>DRAWN BY.:</b> | ETS/AO/IA/JG  |
| <b>JOB NO.:</b>   | PHX21-5038-00 |

**SHEET**

1F0-E617

PLOT DATE: 06/19/2025

**PROJECT ALPHA**  
1F0 - MAIN MANUFACTURING BUILDING  
EAST PECOS ROAD  
QUEEN CREEK, ARIZONA 85140



# WARE MALCOMB

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